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METHODS AND APPARATUSES FOR OPERATING A MEMORY DEVICE

Abstract

Example memory devices, memory systems, and methods for managing over-programming in flash memory are disclosed. In one example, a method of operating a memory device includes, before erasing data in a memory block of the memory device, programing memory cells in the memory block to a pre-programmed state, and identifying a threshold voltage of the memory cells in the pre-programmed state. The method further includes, after erasing the data in the memory block, programing memory cells in the memory block using at least one program pulse determined based on the threshold voltage.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/CN2024/080907, filed on Mar. 11, 2024, the disclosure of which is hereby incorporated by reference in its entirety.

TECHNICAL FIELD
[0002] This present disclosure generally relates to the field of semiconductor technology, and more particularly, to systems and methods for managing over-programming in memory devices.
BACKGROUND
[0003] Flash memory is a low-cost, high-density, nonvolatile solid-state storage medium that can be electrically erased and reprogrammed. Flash memory includes NOR flash memory and NAND flash memory. Various operations can be performed by flash memory, for example, program (write) and erase operations, to change the threshold voltage of each memory cell to a respective level. For NAND flash memory, an erase operation can be performed at the block level, a program operation can be performed at the page level, and a read operation can be performed at the page level.
SUMMARY
[0004] The present disclosure relates to memory devices, memory systems, and methods for managing over-programming of flash memory. In an example, a method of operating a memory device includes, before erasing data in a memory block of the memory device, programing memory cells in the memory block to a pre-programmed state, and identifying a threshold voltage of the memory cells in the pre-programmed state. The method further includes, after erasing the data in the memory block, programing memory cells in the memory block using at least one program pulse determined based on the threshold voltage.
[0005] While generally described as computer-implemented software embodied on tangible media that processes and transforms the respective data, some or all of the aspects may be computer-implemented methods or further included in respective systems or other devices for performing this described functionality. The details of these and other aspects and implementations of the present disclosure are set forth in the accompanying drawings and the description below. Other features, objects, and advantages of the disclosure will be apparent from the description and drawings, and from the claims.

Description

BRIEF DESCRIPTION OF DRAWINGS
[0006] FIG. **1** illustrates an example of a schematic diagram of a memory device including peripheral circuits, according to some aspects of the present disclosure.
[0007] FIG. **2** illustrates an example of a side view of cross-sections of a memory cell array including NAND memory strings, according to some aspects of the present disclosure.
[0008] FIG. **3** illustrates some example peripheral circuits, according to some aspects of the present disclosure.
[0009] FIG. **4**A illustrates an example program pulse, according to some aspects of the present disclosure.
[0010] FIG. **4**B illustrates an example incremental step pulse programming scheme, according to some aspects of the present disclosure.
[0011] FIG. **5**A illustrates a schematic diagram of an example threshold voltage distribution of memory cells in a memory device, according to some aspects of the present disclosure.
[0012] FIG. **5**B illustrates a schematic diagram of an example threshold voltage distribution of memory cells in a memory device after a pre-program operation, according to some aspects of the

present disclosure.
[0013] FIG. **6** illustrates an example schematic diagram of a page buffer of a memory device, according to some aspects of the present disclosure.
[0014] FIGS. **7**A-**7**D illustrate schematic diagrams of example threshold voltage distributions of memory cells in a memory device, according to some aspects of the present disclosure.
[0015] FIG. **8** illustrates an example flow chart of performing a program operation after an erase operation, according to some aspects of the present disclosure.
[0016] FIG. **9** illustrates a block diagram of an example system having a memory device, according to some aspects of the present disclosure.
[0017] FIG. **10**A illustrates a diagram of a memory card having a memory device, according to some aspects of the present disclosure.
[0018] FIG. **10**B illustrates a diagram of a solid-state drive (SSD) having a memory device, according to some aspects of the present disclosure.
[0019] Like reference numbers and designations in the various drawings indicate like elements.

DETAILED DESCRIPTION

[0020] A memory device, such as a NAND memory device, can store data by changing the logic state of memory cells in the memory device. In a program operation, a memory cell changes from an erased state to a programmed state. Specifically, by applying a program voltage to a control gate of the memory cell, electron charges can tunnel through a dielectric layer and remain trapped in a storage layer of the memory device. The charges trapped in the storage layer can increase the threshold voltage of the memory cell. The increased threshold voltage can represent the programmed state of the memory cell.
[0021] In a NAND memory device, data that occupy the memory cells need to be erased before new data can be stored. That is, the memory device needs to perform an erase operation before performing a next program operation on the memory cells, which is generally referred to as a program-erase cycle (P/E cycle). In some cases, when a memory cell has undergone multiple P/E cycles, the memory cell can be programmed by the program voltage to a threshold voltage higher than the intended value (i.e., the threshold voltage of a fresh memory cell programmed using the same program voltage). In such cases, the memory cells can be “over-programmed” after undergoing multiple P/E cycles. The over-programming issue can occur due to several factors, such as excessive charges trapped in the storage layer, degradation of the insulating oxide layer, wear-and-tear of the memory cells. The over-programming issue can affect the reliability of the memory device, such as causing read failures and disturbing adjacent memory cells.
[0022] This present disclosure provides techniques to adjust the program voltage for memory cells that have undergone multiple P/E cycles. In some implementations, before erasing data in a memory block, the memory cells in the memory block are first programmed to a pre-programmed state. One or more verify operations can identify a threshold voltage that represents the pre-programmed state. Based on the identified threshold voltage representing the pre-programmed state, the memory device can adjust the program voltage for a next program operation on the memory block. For example, if the identified threshold voltage is higher than a pre-set voltage, it indicates that the memory cells in the memory block will likely be over-programmed if a pre-defined program voltage is applied in the next program operation. As such, the program voltage for the next program operation should be adjusted to be lower than the pre-defined program voltage to mitigate the over-programming issue.
[0023] FIG. **1** illustrates an example of a schematic circuit diagram of a memory device **100** including peripheral circuits, according to some aspects of the present disclosure. The memory device **100** can include a memory cell array **101** and peripheral circuits **102** coupled to the memory cell array **101**. The memory cell array **101** can be a NAND Flash memory cell array in which memory cells **106** are provided in the form of an array of NAND memory strings **108** each extending vertically above a substrate (not shown in FIG. **1**). In some implementations, each

NAND memory string **108** includes a plurality of memory cells **106** coupled in series and stacked vertically. Each memory cell **106** can hold a continuous, analog value, such as an electrical voltage or charge that depends on the number of electrons trapped within a storage layer of the memory cell **106**. The logic state (i.e., data) of each memory cell **106** in the block **104** can be determined based on the threshold voltage V.sub.th of the memory cell **106**. Each memory cell **106** can be a floating gate type memory cell including a floating-gate transistor, or a charge trap type memory cell including a charge-trap transistor.

[0024] In some implementations, each memory cell **106** is a single-level cell (SLC) with two possible memory states that can store one bit of data. For example, the first memory state “0” can correspond to a first range of voltages, and the second memory state “1” can correspond to a second range of voltages. In some implementations, each memory cell **106** is a multi-level cell (MLC) that is capable of storing more than one bit of data in more than two memory states. For example, the MLC can store two bits per cell, three bits per cell (also known as triple-level cell (TLC)), or four bits per cell (also known as a quad-level cell (QLC)). Each MLC can be programmed to support a range of possible nominal storage values. In one example, if each MLC stores two bits of data, then the MLC can be programmed to one of three possible programming levels from an erased state by writing one of three possible nominal storage values to the cell. A fourth nominal storage value can be used for the erased state.

[0025] As shown in FIG. **1**, each NAND memory string **108** can include a source select gate (SSG) **110** at its source end and a drain select gate (DSG) **112** at its drain end. The SSG **110** and the DSG **112** can be configured to activate selected NAND memory strings **108** (columns of the array) during read and program operations. In some implementations, the sources of NAND memory strings **108** in the same block **104** are coupled through a same source line (SL) **114**, e.g., a common SL. In other words, NAND memory strings **108** in the same block **104** have an array common source (ACS), according to some implementations. The DSG **112** of each NAND memory string **108** is coupled to a respective bit line **116** from which data can be read or written via an output bus (not shown), according to some implementations. In some implementations, each NAND memory string **108** is configured to be selected or deselected by applying a select voltage (e.g., above the threshold voltage of the transistor having the DSG **112**) or a deselect voltage (e.g., 0 V) to the respective DSG **112** through one or more DSG lines **113**, and/or by applying a select voltage (e.g., above the threshold voltage of the transistor having the SSG **110**) or a deselect voltage (e.g., 0 V) to the respective SSG **110** through one or more SSG lines **115**.

[0026] As shown in FIG. **1**, NAND memory strings **108** can be organized into multiple blocks **104**, each of which can have a common source line **114** coupled to the ACS. In some implementations, each block **104** can serve as a basic data unit for erase operations, such that memory cells **106** on the same block **104** are erased at the same time. To erase memory cells **106** in a selected block **104**, the source lines **114** coupled to the selected block **104** and unselected blocks in the same plane can be biased with an erase voltage (Vers). For example, the erase voltage can be a high positive voltage (e.g., 20 V or more). In some implementations, an erase operation can be performed at a half-block level, a quarter-block level, or a level having any suitable number of blocks or fractions of a block.

[0027] The memory cells **106** of adjacent NAND memory strings **108** can be coupled through word lines **118**. The word line **118** can select which row of memory cells **106** is affected by read and program operations. In some implementations, the memory cell **106** is a SLC, and each word line **118** is coupled to a page **120** of memory cells **106**, which is the basic data unit for program operations. If the memory cell **106** is an MLC that stores two bits of data per cell, each word line **118** can correspond to two pages. If memory cell **106** is a TLC, each word line **118** can correspond to three pages. If memory cell **106** is a QLC, each word line **118** can correspond to four pages. The size of a page **120** in bits is associated with the number of NAND memory strings **108** coupled by word line **118** in a block **104**. Each word line **118** can include a gate line coupled to a plurality of control gates (gate electrodes) of a plurality of memory cells **106** in the respective page **120**. Example word lines shown in FIG. **1** include dummy WL, WL**1**, WL**2**, WL**3**, WL**4**, and WL**5** that are between one or more DSG lines **113** and one or more SSG lines **115**.

[0028] FIG. **2** illustrates an example of a side view of cross-sections of a memory cell array **101** including NAND memory strings **108**, according to some aspects of the present disclosure. As shown in FIG. **2**, the NAND memory string **108** can extend vertically through a memory stack **204** above a substrate **202**. The substrate **202** can include silicon (e.g., single crystalline silicon), silicon germanium (SiGe), gallium arsenide (GaAs), germanium (Ge), silicon on insulator (SOI), germanium on insulator (GOI), or any other suitable materials.

[0029] The memory stack **204** can include pairs of interleaved gate conductive layers **206** and gate-to-gate dielectric layers **208**. The quantity of the pairs of the interleaved gate conductive layers **206** and gate-to-gate dielectric layers **208** in a memory stack **204** can determine the quantity of memory cells **106** in memory cell array **101**. The gate conductive layer **206** can include conductive materials including, but not limited to, one or more of tungsten (W), cobalt (Co), copper (Cu), aluminum (Al), polysilicon, doped silicon, or silicide. In some implementations, each gate conductive layer **206** includes a metal layer, such as a tungsten layer. In some implementations, each gate conductive layer **206** includes a doped polysilicon layer. Each gate conductive layer **206** can include control gates surrounding the memory cells **106**, the DSG **112**, or the SSG **110**, and can extend laterally as the DSG line **113** at the top of memory stack **204**, the SSG line **115** at the bottom of memory stack **204**, or the word lines **118** between the DSG line **113** and the SSG line **115**.

[0030] Peripheral circuits **102** can be coupled to the memory cell array **101** through bit lines **116**,**126**,**136**, word lines **118**, source lines **114**, SSG lines **115**, and DSG lines **113**. The peripheral circuits **102** can include any suitable analog, digital, and mixed-signal circuits for facilitating the operations of the memory cell array **101** by applying and sensing voltage signals and/or current signals to and from each target memory cell **106** through bit lines **116**, word lines **118**, source lines **114**, SSG lines **115**, and DSG lines **113**. The peripheral circuits **102** can include various types of peripheral circuits formed using metal-oxide-semiconductor (MOS) technologies. For example, FIG. **3** illustrates some example peripheral circuits, according to some aspects of the present disclosure. The example peripheral circuits include a page buffer/sense amplifier **304**, a column decoder/bit line driver **306**, a row decoder/word line driver **308**, a voltage generator **310**, control logic **312**, registers **314**, an interface **316**, and a data bus. In some examples, additional peripheral circuits not shown in FIG. **3** may be included as well.

[0031] The page buffer/sense amplifier **304** can be configured to read and program (write) data from and to memory cell array **101** according to the control signals from control logic **312**. In an example, the page buffer/sense amplifier **304** may store one page of program data (write data) to be programmed into one page **120** of the memory cell array **101**. In another example, the page buffer/sense amplifier **304** may perform program verify operations to ensure that the data has been properly programmed into memory cells **106** coupled to selected word lines **118**. In still another example, the page buffer/sense amplifier **304** may also sense the low power signals from the bit line **116** that represents a data bit stored in memory cell **106**, and amplify the small voltage swing to recognizable logic levels in a read operation. The column decoder/bit line driver **306** can be configured to be controlled by the control logic **312** and select one or more NAND memory strings **108** by applying bit line voltages generated from the voltage generator **310**.

[0032] The row decoder/word line driver **308** can be configured to be controlled by the control logic **312** and select/deselect blocks **104** of the memory cell array **101** and select/deselect word lines **118** of the block **104**. The row decoder/word line driver **308** can be further configured to drive word lines **118** using word line voltages generated from the voltage generator **310**. In some implementations, the row decoder/word line driver **308** can also select/deselect and drive SSG lines **115** and DSG lines **113**. As described below in detail, the row decoder/word line driver **308** is configured to apply a program voltage to selected word line **118** in a program operation on memory

cell **106** coupled to selected word line **118**.
[0033] The voltage generator **310** can be configured to be controlled by the control logic **312** and generate the word line voltages (e.g., read voltage, program voltage, pass voltage, local voltage, verify voltage, etc.), bit line voltages, and source line voltages to be supplied to the memory cell array **101**.
[0034] The control logic **312** can be coupled to each peripheral circuit described above and configured to control operations of each peripheral circuit. The registers **314** can be coupled to the control logic **312** and include status registers, command registers, and address registers for storing status information, command operation codes (OP codes), and command addresses for controlling the operations of each peripheral circuit.
[0035] The interface **316** can be coupled to the control logic **312** and act as a control buffer to buffer and relay control commands received from a host (not shown) to the control logic **312** and status information received from the control logic **312** to the host. The interface **316** can also be coupled to the column decoder/bit line driver **306** via a data bus, and act as a data input/output (I/O) interface and a data buffer to buffer and relay data to and from the memory cell array **101**.
[0036] Referring back to FIG. **1**, in some implementations, a NAND flash memory can be configured to operate in a single-level cell (SLC) mode. In some implementations, the memory cell **106** can be in an erased state ER or a programmed state P**1**. Initially, the memory cells **106** in the block **104** can be reset to the erased state ER, by removing the trapped electron charges in the storage layer of the memory cell **106**. The trapped charge carriers can be removed by implementing a negative voltage difference between the control gate of the memory cell **106** and the channel of the memory cell. In some implementations, the negative voltage difference can be implemented by setting the control gate of memory cell **106** to the ground, and applying a high positive voltage (i.e., an erase voltage V.sub.ers) to the ACS. At the erased state ER ("state ER" or logic "0"), the threshold voltage V.sub.th of the memory cell **106** can be reset to the lowest value.
[0037] The memory cell **106** can be programmed from state ER to a programmed state P**1** ("state P**1**" or logic "0"). In some implementations, during a program operation, a program pulse **410** as shown in FIG. **4**A can be applied to the word line **118** coupled to the memory cell. The program pulse **410** can implement a positive voltage difference between the control gate and channel of the memory cell **106**. For example, the program pulse **410** can have a program voltage V.sub.pgm (e.g., a positive voltage between 10 V and 30 V), and can have a pulse length (e.g., a time duration between 1 μs to 30 μs) during which the program voltage is applied. During the program operation, the corresponding bit line **116** connecting to the memory cell **106** can be connected to the ground. As a result, electron charges can be injected into the storage layer of the memory cell **106**, thereby increasing the threshold voltage V.sub.th of the memory cell **106**. Accordingly, the memory cell **106** is programmed to state P**1**.
[0038] In some implementations, to increase storage capacity, a NAND flash memory can also be configured to operate in a multi-level cell (MLC) mode, a triple-level cell (TLC) mode, a quad-level cell (QLC) mode, or a combination of any of these modes. In the SLC mode, as mentioned above, a memory cell stores 1 bit of data and has two logic states, logic {1 and 0}, i.e., states ER and P**1**. In the MLC mode, a memory cell stores 2 bits of data, and has four logic states, logic {11, 10, 01, and 00}, i.e., states ER, P**1**, P**2**, and P**3**. In the TLC mode, a memory cell stores 3 bits of data, and has eight logic states, logic {111, 110, 101, 100, 011, 010, 001, 000}, i.e., states ER, and states P**1**-P**7**. In the QLC mode, a memory cell stores 4 bits of data and has 16 logic states, logic {1111, 1110, 1101, 1100, 1011, 1010, 1001, 1000, 0111, 0110, 0101, 0100, 0011, 0010, 0001, 0000}, i.e., states ER, and states P**1**-P**15**.
[0039] In some implementations, memory cells of MLCs, TLCs and QLCs can be programmed to different programmed states using an incremental step pulse programming (ISPP) scheme **420**, as shown in FIG. **4**B. The ISPP scheme **420** can include a plurality of program pulses **430**, **440**, **450**, **460**, **470**, and the memory device can apply several program pulses to program the memory cell

106 to a certain programmed state. Each program pulse can have a program voltage V.sub.pgm (e.g., a positive voltage between 10 V and 30 V), and can have a pulse length (e.g., a time duration between 1 μs to 30 μs) during which the program voltage is applied. In some implementations, a starting program pulse **430** can have a program voltage V.sub.pgm_start, and the program voltages of the following program pulses **440**, **450**, **460**, **470** each increment by voltage ΔV.sub.pgm. The pulse length of the program pulses **430**, **440**, **450**, **460**, **470** can be the same.
[0040] In some implementations, the ISPP scheme **420** further includes at least one verify voltage **480** after a program pulse (e.g., the program pulse **430**). The memory device can perform a verify operation using the verify voltage **480** to determine whether the memory cell **106** has been properly programmed to a target programmed state. For example, the memory device can apply the verify voltage **480** to the word line of the selected memory cell, and a pass voltage V.sub.pass to word lines of unselected memory cells that are coupled to the same bit line as the selected memory cell. If the verify operation indicates that the memory cell **106** has not been properly programmed to the target programmed state, the memory device can apply another program pulse (e.g., the program pulse **440**) to the memory cell **106**. In some implementations, the ISPP scheme **420** can include a verify voltage following each program pulse **430**, **440**, **450**, **460**, **470**.
[0041] For example, the TLC cells can be programmed using the ISPP scheme **420** from state ER with a lowest threshold voltage, to state P**1** with a higher threshold voltage, or to state P**7** with a highest threshold voltage. Each logic state of the memory cells can correspond to a specific range of threshold voltage V.sub.th of the memory cells, where the threshold voltage V.sub.th distribution of each state can be represented by a probability density.
[0042] FIG. **5**A illustrates a schematic diagram of an example threshold voltage distribution of memory cells in a memory device, according to some aspects of the present disclosure. In some implementations, a memory device can perform a program operation on a portion of the memory cells in a memory block to a programmed state (e.g., state P**1**), while the remaining cells are in an erased state (i.e., state ER). In the example of diagram **510**, for a memory device operating in a SLC mode, the threshold voltages V.sub.th of the memory cells in state ER are in a first voltage range, as shown by the V.sub.th distribution curve **512**. The threshold voltages of the memory cells in state P**1** are in a second rage, as shown by V.sub.th distribution curve **514**. For another example, in a memory device operating in a TLC mode, the memory device can program the memory cells to multiple programmed states P**1** to P**7**, while the threshold voltages of the memory cells in each programmed state are in a respective voltage range.
[0043] Before performing another program operation, the memory device needs to first perform an erase operation to erase the data stored in the memory cells. In some implementations, the memory device can perform a pre-program operation on the memory cells before performing the erase operation. The pre-program operation can program the memory cells (including memory cells in the erased state and in one or more programmed states) in a memory block to be erased to a pre-programmed state. For example, in a pre-program operation of an SLC memory device, memory cells in state ER can be programmed to the programmed state P**1**. For another example, in a pre-program operation of a TLC memory device, memory cells in state ER can be programmed to the pre-programmed state similar to state P**5**, and memory cells in one or more of the other programmed states P**1**-P**7** can also be programmed to the pre-programmed state. The pre-program operation can help prevent the memory cells in the erased state or lower programmed states from being over-erased by the later erase operation.
[0044] FIG. **5**B illustrates a schematic diagram of an example threshold voltage distribution of memory cells in a memory device after a pre-program operation, according to some aspects of the present disclosure. In some implementations, before performing an erase operation on a memory block, a pre-program voltage can be applied to the control gates of the memory cells in the memory block, and the memory cells can be programmed to a pre-programmed state. The threshold voltages of the memory cells in the pre-programmed state are within a range, as shown by V.sub.th

distribution curve **522**. In some implementations, the pre-programmed voltage can be the same as a pre-defined program voltage for a SLC memory device, or can be a pre-defined program voltage for one of the programmed states of a MLC, TLC, or QLC memory device.

[0045] In some implementations, after programming the memory cells to be erased to the pre-programmed state, the memory device can perform one or more verify operations to identify the threshold voltage (V.sub.th) that represents the V.sub.th distribution of the pre-programmed state. By applying one or more of the read reference voltages (e.g., V.sub.R1, V.sub.R2, V.sub.R3, V.sub.R4) to the control gate of the memory cells, the memory device can determine a voltage that represents the V.sub.th distribution of the memory cells in the pre-programmed state (e.g., the left x-intercept of the V.sub.th distribution curve **522**). For example, a quantity of memory cells in an on-state (i.e., a fail bit count, FBC) when V.sub.R1 is applied to the control gates of the memory cells is recorded as N.sub.1. N.sub.1 can represent the quantity of the memory cells whose threshold voltage is lower than V.sub.R1. Similarly, a quantity of memory cells in an on-state when V.sub.R2, V.sub.R3 or V.sub.R4 (V.sub.R1>V.sub.R2>V.sub.R3>V.sub.R4) is applied to the control gates of the memory cells is recorded as N.sub.2, N.sub.3, N.sub.4, respectively (N.sub.1>N.sub.2>N.sub.3>N.sub.4).

[0046] In some implementations, the verify operations can be performed based on read reference voltages from higher to lower. For example, the verify operation using V.sub.R1 is performed first, and then the verify operations using smaller read reference voltages V.sub.R2, V.sub.R3, V.sub.R4 are performed respectively. The threshold voltage can be identified by comparing the FBCs of the read reference voltages to a pre-set threshold. For example, if N.sub.1, N.sub.2 and N.sub.3 are greater than the pre-set threshold, and N.sub.4 is less than the pre-set threshold, V.sub.R4 is determined as V.sub.th that represents the V.sub.th distribution of the pre-programmed state. For another example, if N.sub.1 and N.sub.2 are greater than the pre-set threshold, and N.sub.3 is less than the pre-set threshold, V.sub.R3 is determined as the V.sub.th that represents the V.sub.th distribution of the pre-programmed state, and the verify operation using V.sub.R4 can be skipped.

[0047] In other implementations, the verify operations can be performed based on read reference voltages (e.g., V.sub.R1′, V.sub.R2′, V.sub.R3′ and V.sub.R4′, where V.sub.R1′<V.sub.R2′<V.sub.R3′<V.sub.R4′) from lower to higher. V.sub.R1′ is lower than the left x-intercept of the V.sub.th distribution curve **522**. For example, the verify operation using V.sub.R1′ is performed first, and then the verify operations using larger read reference voltages V.sub.R2′, V.sub.R3′, V.sub.R4′ are performed respectively. The threshold voltage can be identified by comparing the FBCs of the read reference voltages to a pre-set threshold. For example, if N.sub.1′, N.sub.2′ and N.sub.3′ are less than the pre-set threshold, and N.sub.4′ is greater than the pre-set threshold, V.sub.R4′ is determined as the V.sub.th that represents the V.sub.th distribution of the pre-programmed state. For another example, if N.sub.1′ and N.sub.2′ are less than the pre-set threshold, and N.sub.3′ is greater than the pre-set threshold, V.sub.R3′ is determined as the V.sub.th that represents the V.sub.th distribution of the pre-programmed state, and the verify operation using V.sub.R4′ can be skipped.

[0048] FIG. **6** illustrates example schematic diagram of a page buffer of a memory device, according to some aspects of the present disclosure. FIG. **6** can be described with regard to the page buffer/sense amplifier **304** of FIG. **3**. Not all of the depicted components may be used, however, and one or more implementations may include additional components not shown in the figure. Variations in the arrangement and types of the components may be made without departing from the spirit or scope of the claims as set forth herein. Additionally, different or fewer components may be provided.

[0049] The page buffer includes a plurality of buffer structures **658**. In some examples, a page buffer structure **658** connects with the memory string **108** through the bit line **116** of FIG. **1**. In some examples, each page buffer of the plurality of page buffers connects with a corresponding memory string through a bit line.

[0050] As shown in FIG. **6**, the page buffer structure **658** can include a sensing node (aka., sense-out node, or SO) **650**, a precharge path **652**, a SO discharge path **654**, an L latch **682**, and a sense latch **656**. The page buffer structure **658** can further includes a cache latch and one or more data latches.
[0051] As described with reference to FIG. **5**B, during a verify operation, the row decoder/word line driver **308** can transfer a read reference voltage V.sub.R1, V.sub.R2, V.sub.R3, or V.sub.R4 to a selected word line, and a pass voltage V.sub.pass to an unselected word line. The page buffer can sense current flowing through the bit line **116** that reflects the logic state (i.e., data) of the memory cell **106** and amplify small signal to a measurable magnification. For example, the page buffer structure **658** can precharge the SO **650** through the precharge path **652** by a control circuit (e.g., control logic **312**), and can sense at the SO **650** whether a selected memory cell is turned on or off. Specifically, if the threshold voltage of the selected memory cell is lower than the read reference voltage, the selected memory cell should be switched on to form a conductive path in the channel. A charge charged at the SO **650** may be quickly discharged through the channel of the memory string (i.e., through the SO discharge path **654**), and the voltage at SO **650** can drop to a low level after a develop period. A logic value 1 may be latched to the sense latch **656**. Alternatively, if the threshold voltage of the selected memory cell is higher than the read reference voltage V.sub.R1, the selected memory cell should remain switched off. The charge charged at the SO **650** may not be quickly discharged through the channel of the cell string (i.e., through the SO discharge path **654**), and the voltage at SO **650** will remain at a high level after the develop period. The page buffer structure **658** can obtain the result of the verify operation by measuring the state of the SO **650** after the develop period. A logic value 0 may be latched to the sense latch **656**.
[0052] In some implementations, after performing verify operations on all target memory cells (e.g., all the pre-programmed memory cells in a memory block, or all the pre-programmed memory cells coupled to the same word line) using a read reference voltage, the sum of the logic value 1 latched to the sense latch **656** is identified as the FBC of the read reference voltage. For example, after performing a set of verify operations based on a set of read reference voltages V.sub.R1, V.sub.R2, V.sub.R3, and V.sub.R4, each read reference voltage can have a corresponding FBC (i.e., N.sub.1, N.sub.2, N.sub.3, N.sub.4 as described with reference to FIG. **5**B).
[0053] FIG. **7**A-**7**D schematic diagrams of example threshold voltage distributions of memory cells in a memory device, according to some aspects of the present disclosure. Due to multiple factors such as excessive charges trapped in the storage layer, degradation of the insulating oxide layer and wear-and-tear, the memory cells after multiple cycles (e.g., after 500 or 1000 program-erase cycles) can react differently to the same program voltage or the same erase voltage, as compared to memory cells in fresh cycles (e.g., the initial 500 or 1000 program-erase cycles). As shown in FIG. **7**A, curve **710** represents a threshold voltage (V.sub.th) distribution of memory cells in fresh cycles in an erased state. Curve **720** represents a V.sub.th distribution of memory cells after multiple cycles in an erased state. In some implementations, curve **720** drifts to the right of curve **710**. That is, by applying the same erase voltage, the V.sub.th of memory cells after multiple cycles in the erased state can be higher than the V.sub.th of the memory cells in fresh cycles in the erased state. Curve **712** represents a V.sub.th distribution of memory cells in fresh cycles in a programmed state. Curve **722** represents a V.sub.th distribution of memory cells after multiple cycles in a programmed state. In some implementations, curve **722** drifts to the right of curve **712**. That is, by applying the same program voltage, the V.sub.th of memory cells after multiple cycles in the programmed state can be higher than the V.sub.th of the memory cells in fresh cycles in the programmed state. As such, the memory cells that have undergone multiple cycles can be over-programmed beyond the intended V.sub.th level, which can affect the reliability of the memory device.
[0054] In some implementations, as discussed with reference to FIG. **5**B, prior to erasing data in a memory block, the memory cells in the memory block can first be pre-programmed to a pre-programmed state **522**. For example, the pre-program operation can program the memory cells

(including memory cells in the erased state and in one or more programmed states) in a memory block to be erased to the same pre-programmed state. After the pre-program operation, the memory device can perform one or more verify operations based on one or more read reference voltages to identify the threshold voltage of the memory cells in the pre-programmed state. As shown in FIG. **7**B, curve **714** represents a V.sub.th distribution of memory cells in fresh cycles in the pre-programmed state. Curve **724** represents a V.sub.th distribution of memory cells after multiple cycles in the pre-programmed state. In some implementations, curve **724** drifts to the right of curve **714**. That is, under the same pre-program voltage, the V.sub.th of memory cells after multiple cycles in the pre-programmed state can be higher than the V.sub.th of memory cells in fresh cycles in the pre-programmed state.

[0055] In some implementations, a program pulse for a next program operation can be adjusted based on the identified V.sub.th that represents curve **724**. As discussed with reference to FIG. **4**A, in a memory device operating in an SLC mode, the memory device can adjust the program pulse **410** for the next program operation, based on the threshold voltage (e.g., V.sub.R4) identified by the verify operations that can represent curve **724**. For example, the program voltage V.sub.pgm of the program pulse **410** can be adjusted to be lower than a pre-defined program voltage V.sub.pgm_initial. The difference between the V.sub.pgm_initial and V.sub.pgm can be positively correlated to the identified V.sub.th that represents curve **724**. For instance, if the V.sub.th that represents curve **724** is lower than V.sub.R1, the program voltage V.sub.pgm can remain the same as V.sub.pgm_initial. If the V.sub.th that represents curve **724** is higher than V.sub.R1 but lower than V.sub.R2, the program voltage V.sub.pgm can be V.sub.pgm_initial−ΔV.sub.1 (e.g., ΔV.sub.1=0.1V). If the V.sub.th that represents curve **724** is higher than V.sub.R2 but lower than V.sub.R3, the program voltage V.sub.pgm can be V.sub.pgm_initial−ΔV.sub.2, and ΔV.sub.2 is greater than ΔV.sub.1 (e.g., ΔV.sub.1=0.2V). For another example, the pulse length of the program pulse **410** can be adjusted to be shorter than a pre-defined pulse length.

[0056] In some implementations, as discussed with reference to FIG. **4**B, in a memory device operating in an MLC, TLC or QLC mode, the memory device can adjust the ISPP scheme **420** for the next program operation based on the V.sub.th that represents curve **724**. For example, the program voltage V.sub.pgm_start of the starting program pulse **430** can be adjusted to be lower than a pre-defined program voltage V.sub.pgm_start_initial, while the incremental voltage ΔV.sub.pgm of the ISPP scheme **420** can remain unchanged. For instance, if the V.sub.th that represents curve **724** is lower than V.sub.R1, the program voltage V.sub.pgm_start can remain the same as V.sub.pgm_start_initial. If the V.sub.th that represents curve **724** is higher than V.sub.R1 but lower than V.sub.R2, the program voltage V.sub.pgm_start can be V.sub.pgm_start_initial−ΔV.sub.1 (e.g., ΔV.sub.1=0.1V). If the V.sub.th that represents curve **724** is higher than V.sub.R2 but lower than V.sub.R3, the program voltage V.sub.pgm_start can be V.sub.pgm_start_initial−ΔV.sub.2, and ΔV.sub.2 is greater than ΔV.sub.1 (e.g., ΔV.sub.1=0.2V). For another example, the pulse length of the program pulses **430**, **440**, **450**, **460**, **470** can be adjusted to be shorter than a pre-defined pulse length.

[0057] In some implementations, the adjusted program voltage or the adjusted pulse length can be stored in a register (e.g., one of the registers **314** in FIG. **3**) or a SRAM (not shown) of the memory device **100**. As such, the memory device can directly apply the adjusted program voltage or the adjusted pulse length in subsequent program operations.

[0058] As shown in FIG. **7**C, after the pre-program operation, the memory device performs an erase operation on the memory cells. Curve **716** represents a V.sub.th distribution of memory cells in fresh cycles in the erased state, similar to curve **710**. Curve **726** represents a V.sub.th distribution of memory cells after multiple cycles in the erased state, similar to curve **720**.

[0059] As shown in FIG. **7**D, in some implementations, the memory device applies the adjusted program voltage in a program operation subsequent to the erase operation. For example, the memory device can apply the pre-defined program voltage V.sub.pgm_initial to program memory cells in fresh cycles. As such, the memory cells in the erased state as represented by V.sub.th distribution curve **718** are programmed to the programmed state as represented by V.sub.th distribution curve **719**. For another example, the memory device can apply the adjusted program voltage V.sub.pgm to program memory cells that have undergone multiple cycles. As such, the memory cells in the erased state as represented by V.sub.th distribution curve **728** are programmed to the programmed state as represented by V.sub.th distribution curve **729**. The V.sub.th distribution curve **729** is close to the V.sub.th distribution curve **719**, which indicates that the memory cells after multiple cycles can be programmed to the intended V.sub.th level, instead of being over-programmed. As a comparison, if the memory device applies the pre-defined program voltage V.sub.pgm_initial to program memory cells that have undergone multiple cycles, the memory cells can be over-programmed beyond the intended V.sub.th level, as represented by V.sub.th distribution curve **740**.

[0060] FIG. **8** illustrates an example flow chart of performing a program operation after an erase operation.

[0061] At operation **802**, a memory device (e.g., the memory device **100** in FIG. **1**) receives an erase command from a memory controller (e.g. the memory controller **906** in FIG. **9**) coupled to the memory device. The erase command can indicate to erase data in one or more memory blocks (e.g., block **104** of FIG. **1**). In some implementations, other than an erase command from a host (e.g., host **908** of FIG. **9**) sent by the memory controller, the erase operation can be triggered by operations such as garbage collection, bad block management, wear-leveling, etc.

[0062] At operation **804**, the memory device can perform a pre-program operation on the memory block to be erased. For example, the memory device can apply a pre-program voltage through the word lines coupled to the memory cells in the memory block. In some implementations, the pre-programmed voltage can be the same as a pre-defined program voltage for a SLC memory device, or can be a pre-defined program voltage for one of the programmed states of a MLC, TLC, or QLC memory device. The memory cells in the memory block, including the memory cells in the erased state (e.g., curve **720** in FIG. **7**A) and in one or more programmed states (e.g., curve **722** in FIG. **7**A), are programmed to a pre-programmed state (e.g., curve **724** in FIG. **7**B).

[0063] At operation **806**, the memory device performs a set of verify operations based on a set of read reference voltages on the memory cells in the pre-programmed state. The verify operations can identify a threshold voltage that represents the pre-programmed state (e.g., the left x-intercept of curve **724**). In some implementations, the verify operations can determine a fail bit count (FBC) for each read reference voltage. The FBC represents the quantity of memory cells with threshold voltage higher than the read reference voltage. In some implementations, the FBC of each read reference voltage is compared to a pre-set threshold. The read reference voltage from the set of read reference voltages that is identified as the threshold voltage of the programmed state has a FBC smaller than the pre-set threshold.

[0064] In some implementations, operation **806** is performed when a quantity of program-erase cycles (P/E cycles) undergone by the memory cells has reached a certain limit. For example, if the quantity of P/E cycles of the memory cells is less than a predetermined threshold (e.g., 500 or 1000 P/E cycles), the memory cells are deemed to be in fresh cycles, where over-programming issue is not severe. Therefore operation **806** can be skipped in method **800**. For another example, if the quantity of P/E cycles of the memory cells is greater than the predetermined threshold, the memory cells are deemed to have undergone multiple cycles, where the over-programming issue can affect the reliability of the memory device. Therefore, operation **806** is performed after pre-programming the memory cells in operation **804**.

[0065] At operation **808**, the memory device performs an erase operation on the memory block. As such, the memory cells in the memory block are set to the erased state.

[0066] At operation **810**, the memory device performs a program operation on the memory block using one or more adjusted program pulses. For example, in response to receiving a program command from the memory controller to write data in the memory block, the memory device can perform a program operation on certain pages of the memory block using the adjusted program pulses, instead of the pre-defined program pulses. In some implementations, the memory device can adjust the program voltage based on the threshold voltage of the programmed state identified in operation **806**. For example, the memory device can adjust the program voltage or the pulse length of a program pulse (e.g., the program pulse **410** in FIG. **4**A) for an SLC. For another example, the memory device can adjust the program voltage of a starting program pulse (e.g., the starting program pulse **430** in FIG. **4**B), or a pulse length of the program pulses (e.g., the program pulse **430**, **440**, **450**, **460**, **470** in FIG. **4**B) of a ISPP scheme (e.g., the ISPP scheme **420** in FIG. **4**B) for an MLC, TLC or QLC.

[0067] FIG. **9** illustrates a block diagram of an example system **900** having a memory device, according to some aspects of the present disclosure. System **900** can be a mobile phone, a desktop computer, a laptop computer, a tablet, a vehicle computer, a gaming console, a printer, a positioning device, a wearable electronic device, a smart sensor, a virtual reality (VR) device, an argument reality (AR) device, or any other suitable electronic devices having storage therein. As shown in FIG. **6**, system **900** can include a host **908** and a memory system **902** having one or more memory devices **904** and a memory controller **906**. Host **908** can be a processor of an electronic device, such as a central processing unit (CPU), or a system-on-chip (SoC), such as an application processor (AP). Host **908** can be configured to send or receive data to or from memory devices **904**.

[0068] Memory device **904** can be any memory device disclosed in the present disclosure. Memory controller **906** is coupled to memory device **904** and host **908** and is configured to control memory device **904**, according to some implementations. Memory controller **906** can manage the data stored in memory device **904** and communicate with host **908**. In some implementations, memory controller **906** is designed for operating in a low duty-cycle environment like secure digital (SD) cards, compact Flash (CF) cards, universal serial bus (USB) Flash drives, or other media for use in electronic devices, such as personal computers, digital cameras, mobile phones, etc. In some implementations, memory controller **906** is designed for operating in a high duty-cycle environment SSDs or embedded multi-media-cards (eMMCs) used as data storage for mobile devices, such as smartphones, tablets, laptop computers, etc., and enterprise storage arrays. Memory controller **906** can be configured to control operations of memory device **904**, such as read, erase, and program operations. Memory controller **906** can also be configured to manage various functions with respect to the data stored or to be stored in memory device **904** including, but not limited to bad-block management, garbage collection, logical-to-physical address conversion, wear leveling, etc. In some implementations, memory controller **906** is further configured to process error correction codes (ECCs) with respect to the data read from or written to memory device **904**. Any other suitable functions may be performed by memory controller **906** as well, for example, formatting memory device **904**.

[0069] Memory controller **906** can communicate with an external device (e.g., host **908**) according to a particular communication protocol. For example, memory controller **906** may communicate with the external device through at least one of various interface protocols, such as a USB protocol, an MMC protocol, a peripheral component interconnection (PCI) protocol, a PCI-express (PCI-E) protocol, an advanced technology attachment (ATA) protocol, a serial-ATA protocol, a parallel-ATA protocol, a small computer small interface (SCSI) protocol, an enhanced small disk interface (ESDI) protocol, an integrated drive electronics (IDE) protocol, a Firewire protocol, etc.

[0070] Memory controller **906** and one or more memory devices **904** can be integrated into various types of storage devices, for example, be included in the same package, such as a universal Flash storage (UFS) package or an eMMC package. That is, memory system **902** can be implemented and packaged into different types of end electronic products. In one example as shown in FIG. **10**A, memory controller **906** and a memory device **904** may be integrated into a memory card

1002. Memory card **1002** can include a PC card (PCMCIA, personal computer memory card international association), a CF card, a smart media (SM) card, a memory stick, a multimedia card (MMC, RS-MMC, MMCmicro), an SD card (SD, miniSD, microSD, SDHC), a UFS, etc. Memory card **1002** can further include a memory card connector **1004** coupling memory card **1002** with a host (e.g., host **908** in FIG. **6**). In another example as shown in FIG. **10**B, memory controller **906** and multiple memory devices **904** may be integrated into an SSD **1006**. SSD **1006** can further include an SSD connector **1008** coupling SSD **1006** with a host (e.g., host **908** in FIG. **6**). In some implementations, the storage capacity and/or the operation speed of SSD **1006** is greater than those of memory card **1002**.

[0071] According to one aspect of the present disclosure, a method for operating a memory device is provided. The method includes, before erasing data in a memory block of the memory device, programing memory cells in the memory block to a pre-programmed state, and identifying a threshold voltage of the memory cells in the pre-programmed state. The method further includes, after erasing the data in the memory block, programing memory cells in the memory block using at least one program pulse determined based on the threshold voltage.

[0072] The method can include one or more of the following features.

[0073] In some implementations, the threshold voltage of the memory cells in the pre-programmed state increases with an increase of a quantity of erase operations performed on the memory block.

[0074] In some implementations, programing the memory cells in the memory block to the pre-programmed state includes programming memory cells in an erased state to the pre-programmed state.

[0075] In some implementations, identifying the threshold voltage of the memory cells includes applying a set of read voltages to the memory cells, determining a set of fail bit counts (FBCs) of the memory cells corresponding to the set of the read voltages, and identifying the threshold voltage based on the set of FBCs.

[0076] In some implementations, identifying the threshold voltage of the memory cells further includes comparing the set of FBCs to a threshold. A FBC associated with a read voltage of the set of the read voltages that is identified as the threshold voltage is less than the threshold.

[0077] In some implementations, the memory cells are single-level cells, and the at least one program pulse is a single program pulse. At least one of a program voltage or a pulse length of the single program pulse is determined based on the threshold voltage.

[0078] In some implementations, the program voltage is lower than a pre-defined program voltage. A difference between the pre-defined program voltage and the program voltage is positively correlated to a value of the threshold voltage.

[0079] In some implementations, the memory cells are multi-level cells, and the at least one program pulse are a plurality of program pulses. At least one of a program voltage of a starting program pulse of the plurality of the program pulses or a pulse length of the plurality of program pulses is determined based on the threshold voltage.

[0080] In some implementations, a quantity of erase operations performed on the memory block is greater than a pre-determined threshold.

[0081] According to another aspect of the present disclosure, a memory device is provided. The memory device includes a memory array that includes memory blocks and a peripheral circuit. The peripheral circuit is configured to perform operations including, before erasing data in a memory block, programing memory cells in the memory block to a pre-programmed state, and identifying a threshold voltage of the memory cells in the pre-programmed state. The peripheral circuit is further configured to perform operations including, after erasing data in the memory block, programing memory cells in the memory block using at least one program pulse determined based on the threshold voltage.

[0082] The memory device can include one or more of the following features.

[0083] In some implementations, the threshold voltage of the memory cells in the pre-programmed

state increases with an increase of a quantity of erase operations performed on the memory block.
[0084] In some implementations, programing the memory cells in the memory block to the pre-programmed state includes programming memory cells in an erased state to the pre-programmed state.
[0085] In some implementations, identifying the threshold voltage of the memory cells includes applying a set of read voltages to the memory cells, determining a set of fail bit counts (FBCs) of the memory cells corresponding to the set of the read voltages, and identifying the threshold voltage based on the set of FBCs.
[0086] In some implementations, the memory cells are single-level cells, and the at least one program pulse is a single program pulse. At least one of a program voltage or a pulse length of the single program pulse is determined based on the threshold voltage.
[0087] In some implementations, the program voltage is lower than a pre-defined program voltage. A difference between the pre-defined program voltage and the program voltage is positively correlated to a value of the threshold voltage.
[0088] In some implementations, the memory cells are multi-level cells, and the at least one program pulse are a plurality of program pulses. At least one of a program voltage of a starting program pulse of the plurality of the program pulses or a pulse length of the plurality of program pulses is determined based on the threshold voltage.
[0089] In some implementations, a quantity of erase operations performed on the memory block is greater than a pre-determined threshold.
[0090] According to another aspect of the present disclosure, a memory system is provided. The memory system includes a memory controller and a memory device coupled the memory controller. The memory device is configured to perform operations including, before erasing data in a memory block, programing memory cells in the memory block to a pre-programmed state, and identifying a threshold voltage of the memory cells in the pre-programmed state. The memory device is further configured to perform operations including, after erasing data in the memory block, programing memory cells in the memory block using at least one program pulse determined based on the threshold voltage.
[0091] The memory system can include one or more of the following features.
[0092] In some implementations, the memory controller is configured to send a program command to the memory device. The memory device is further configured to, in response to receiving the program command, perform a program operation on the memory cells using the at least one program pulse. A program voltage of the at least one program pulse is lower than a pre-defined program voltage.
[0093] While this specification contains many specific implementation details, these should not be construed as limitations on the scope of what may be claimed, but rather as descriptions of features that may be specific to particular implementations. Certain features that are described in this specification in the context of separate implementations can also be implemented, in combination, in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations, separately, or in any sub-combination. Moreover, although previously described features may be described as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can, in some cases, be excised from the combination, and the claimed combination may be directed to a sub-combination or variation of a sub-combination.
[0094] It is noted that references in the specification to “one implementation,” “an implementation,” “an example implementation,” “some implementation,” etc., indicate that the implementation described can include a particular feature, structure, or characteristic, but every implementation may not necessarily include the particular feature, structure, or characteristic. Moreover, such phrases do not necessarily refer to the same implementation. Further, when a particular feature, structure or characteristic is described in connection with an implementation, it

would be within the knowledge of a person skilled in the pertinent art to affect such feature, structure, or characteristic in connection with other implementations whether or not explicitly described.

[0095] In general, terminology can be understood at least in part from usage in context. For example, the term "one or more" as used herein, depending at least in part upon context, can be used to describe any feature, structure, or characteristic in a singular sense or can be used to describe combinations of features, structures, or characteristics in a plural sense. Similarly, terms, such as "a," "an," or "the," again, can be understood to convey a singular usage or to convey a plural usage, depending at least in part upon context. In addition, the term "based on" can be understood as not necessarily intended to convey an exclusive set of factors and can, instead, allow for existence of additional factors not necessarily expressly described, again, depending at least in part on context.

[0096] As used herein, the term "substrate" refers to a material onto which subsequent material layers are added. The substrate includes a "top" surface and a "bottom" surface. The top surface of the substrate is typically where a semiconductor device is formed, and therefore the semiconductor device is formed at a top side of the substrate unless stated otherwise. The bottom surface is opposite to the top surface and therefore a bottom side of the substrate is opposite to the top side of the substrate. The substrate itself can be patterned. Materials added on top of the substrate can be patterned or can remain unpatterned. Furthermore, the substrate can include a wide array of semiconductor materials, such as silicon, germanium, gallium arsenide, indium phosphide, etc. Alternatively, the substrate can be made from an electrically non-conductive material, such as a glass, a plastic, or a sapphire wafer.

[0097] As used herein, the term "layer" refers to a material portion including a region with a thickness. A layer has a top side and a bottom side where the bottom side of the layer is relatively close to the substrate and the top side is relatively away from the substrate. A layer can extend over the entirety of an underlying or overlying structure, or can have an extent less than the extent of an underlying or overlying structure. Further, a layer can be a region of a homogeneous or inhomogeneous continuous structure that has a thickness less than the thickness of the continuous structure. For example, a layer can be located between any set of horizontal planes between, or at, a top surface and a bottom surface of the continuous structure. A layer can extend horizontally, vertically, and/or along a tapered surface. A substrate can be a layer, can include one or more layers therein, and/or can have one or more layer thereupon, there above, and/or there below. A layer can include multiple layers. For example, an interconnect layer can include one or more conductive and contact layers (in which contacts, interconnect lines, and/or vertical interconnect accesses (VIAs) are formed) and one or more dielectric layers.

[0098] As used in this disclosure, the term "about" or "approximately" can allow for a degree of variability in a value or range, for example, within 10%, within 5%, or within 1% of a stated value or of a stated limit of a range.

[0099] As used in this disclosure, the term "substantially" refers to a majority of, or mostly, as in at least about 50%, 60%, 70%, 80%, 90%, 95%, 96%, 97%, 98%, 99%, 99.5%, 99.9%, 99.99%, or at least about 99.999% or more.

[0100] Values expressed in a range format should be interpreted in a flexible manner to include not only the numerical values explicitly recited as the limits of the range, but also to include the individual numerical values or sub-ranges encompassed within that range as if each numerical value and sub-range is explicitly recited. For example, a range of "0.1% to about 5%" or "0.1% to 5%" should be interpreted to include about 0.1% to about 5%, as well as the individual values (for example, 1%, 2%, 3%, and 4%) and the sub-ranges (for example, 0.1% to 0.5%, 1.1% to 2.2%, 3.3% to 4.4%) within the indicated range. The statement "X to Y" has the same meaning as "about X to about Y," unless indicated otherwise. Likewise, the statement "X, Y, or Z" has the same meaning as "about X, about Y, or about Z," unless indicated otherwise.

[0101] Particular implementations of the subject matter have been described. Other implementations, alterations, and permutations of the described implementations are within the scope of the following claims as will be apparent to those skilled in the art. While operations are depicted in the drawings or claims in a particular order, such operations are not required be performed in the particular order shown or in sequential order, or that all illustrated operations be performed (some operations may be considered optional), to achieve desirable results. In certain circumstances, multitasking or parallel processing (or a combination of multitasking and parallel processing) may be advantageous and performed as deemed appropriate.
[0102] Moreover, the separation or integration of various system modules and components in the previously described implementations are not required in all implementations, and the described components and systems can generally be integrated together or packaged into multiple products.
[0103] Accordingly, the previously described example implementations do not define or constrain the present disclosure. Other changes, substitutions, and alterations are also possible without departing from the spirit and scope of the present disclosure.

Claims

1. A method of operating a memory device, comprising: before erasing data in a memory block of the memory device: programing memory cells in the memory block to a pre-programmed state; and identifying a threshold voltage of the memory cells in the pre-programmed state; and after erasing the data in the memory block, programing memory cells in the memory block using at least one program pulse determined based on the threshold voltage.
2. The method of claim 1, wherein the threshold voltage of the memory cells in the pre-programmed state increases with an increase of a quantity of erase operations performed on the memory block.
3. The method of claim 1, wherein programing the memory cells in the memory block to the pre-programmed state comprises: programming memory cells in an erased state to the pre-programmed state.
4. The method of claim 1, wherein identifying the threshold voltage of the memory cells comprising: applying a set of read voltages to the memory cells; determining a set of fail bit counts (FBCs) of the memory cells corresponding to the set of the read voltages; and identifying the threshold voltage based on the set of FBCs.
5. The method of claim 4, wherein identifying the threshold voltage of the memory cells comprises: comparing the set of FBCs to a threshold, wherein a FBC associated with a read voltage of the set of the read voltages that is identified as the threshold voltage is less than the threshold.
6. The method of claim 1, wherein the memory cells are single-level cells, wherein the at least one program pulse is a single program pulse, and wherein at least one of a program voltage of the single program pulse or a pulse length of the single program pulse is determined based on the threshold voltage.
7. The method of claim 6, wherein the program voltage is lower than a pre-defined program voltage, wherein a difference between the pre-defined program voltage and the program voltage is positively correlated to a value of the threshold voltage.
8. The method of claim 1, wherein the memory cells are multi-level cells, wherein the at least one program pulse are a plurality of program pulses, and wherein at least one of a program voltage of a starting program pulse of the plurality of the program pulses or a pulse length of the plurality of program pulses is determined based on the threshold voltage.
9. The method of claim 1, wherein a quantity of erase operations performed on the memory block is greater than a pre-determined threshold.
10. A memory device comprising: a memory array comprising memory blocks; and a peripheral circuit configured to perform operations comprising: before erasing data in a memory block:

programing memory cells in the memory block to a pre-programmed state; and identifying a threshold voltage of the memory cells in the pre-programmed state; and after erasing data in the memory block, programing memory cells in the memory block using at least one program pulse determined based on the threshold voltage.

11. The memory device of claim 10, wherein the threshold voltage of the memory cells in the pre-programmed state increases with an increase of a quantity of erase operations performed on the memory block.

12. The memory device of claim 10, wherein programing the memory cells in the memory block to the pre-programmed state comprises: programming memory cells in an erased state to the pre-programmed state.

13. The memory device of claim 10, wherein identifying the threshold voltage of the memory cells comprising: applying a set of read voltages to the memory cells; determining a set of fail bit counts (FBCs) of the memory cells corresponding to the set of the read voltages; and identifying the threshold voltage based on the set of FBCs.

14. The memory device of claim 10, wherein the memory cells are single-level cells, wherein the at least one program pulse is a single program pulse, and wherein at least one of a program voltage of the single program pulse or a pulse length of the single program pulse is determined based on the threshold voltage.

15. The memory device of claim 14, wherein the program voltage is lower than a pre-defined program voltage, wherein a difference between the pre-defined program voltage and the program voltage is positively correlated to a value of the threshold voltage.

16. The memory device of claim 10, wherein the memory cells are multi-level cells, wherein the at least one program pulse are a plurality of program pulses, and wherein at least one of a program voltage of a starting program pulse of the plurality of the program pulses or a pulse length of the plurality of program pulses is calculated based on the threshold voltage.

17. The memory device of claim 10, wherein a quantity of erase operations performed on the memory block is greater than a pre-determined threshold.

18. The memory device of claim 10, wherein the memory device comprises a NAND memory device.

19. A memory system comprising: a memory controller; and a memory device coupled to the memory controller, wherein the memory device is configured to perform operations comprising: before erasing data in a memory block of the memory device: programing memory cells in the memory block to a pre-programmed state; and identifying a threshold voltage of the memory cells in the pre-programmed state; and after erasing the data in the memory block, programing memory cells in the memory block using at least one program pulse determined based on the threshold voltage.

20. The memory system of claim 19, wherein the memory controller is configured to send a program command to the memory device, and wherein the memory device is further configured to: in response to receiving the program command, perform a program operation on the memory cells using the at least one program pulse, wherein a program voltage of the at least one program pulse is lower than a pre-defined program voltage.